

The table below shows the probability distribution of a discrete random variable X , where a is an integer and b is a constant.

x	1	4	a	25
$P(X=x)$	0.1	b	0.35	0.3

(a) Find b . Hence, express $E\left((2\sqrt{X}-5)^2\right)$ in terms of a .

(b) It is given that $\text{Var}(2\sqrt{X}-5) = 7.31$. Find a .

(c) Let C be the event that X is an odd number and D be the event that $X \leq 9$. Are C and D independent? Explain your answer.

(8 marks)

$$a) \quad 0.1 + b + 0.35 + 0.3 = 1$$

$$b = 0.25 \quad \text{IA}$$

$$E[(2\sqrt{X}-5)^2] = E(4x - 20\sqrt{x} + 25)$$

$$= 4(8.6 + 0.35a) - 20(2.1 + 0.35\sqrt{a}) + 25 \quad \text{IM} \quad \text{OR evaluating } E((2\sqrt{X}-5)^2) \text{ directly}$$

$$= 1.4a - 7\sqrt{a} + 17.4 \quad \text{IA}$$

$$b) \quad E(2\sqrt{X}-5)$$

$$= 2(2.1 + 0.35\sqrt{a}) - 5$$

$$= 0.7\sqrt{a} - 0.8$$

$$1.4a - 7\sqrt{a} + 17.4 - (0.7\sqrt{a} - 0.8)^2 = 7.31 \quad \text{IM} \quad \text{OR evaluate using}$$

$$0.91a - 5.88\sqrt{a} + 9.45 = 0 \quad \text{IM}$$

$$\text{Var}(2\sqrt{X}-5) = 4 \text{Var}\sqrt{X}$$

$$= 4(E(X) - (E\sqrt{X})^2)$$

$$\sqrt{a} = \frac{45}{13} \text{ (rejected) or } 3$$

$$\therefore a = 9 \quad \text{IA}$$

$$c) \quad P(C) = 0.1 + 0.35 + 0.3 = 0.75$$

$$P(D) = 0.1 + 0.25 + 0.35 = 0.7$$

$$P(C \cap D) = 0.1 + 0.35 = 0.45$$

$$P(C) \cdot P(D) = 0.525 \neq P(C \cap D)$$

$$\therefore \text{not independent.} \quad \text{I f.t.}$$

IA for either